

OpenStax Calculus Volume 3

Section 5.4 Exercises: Triple Integrals

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In the following exercises, evaluate the triple integrals over the rectangular solid box B .

1. $\iiint_B (2x + 3y^2 + 4z^3) dV$, where $B = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq 2, 0 \leq z \leq 3\}$
2. $\iiint_B (xy + yz + xz) dV$, where $B = \{(x, y, z) | 1 \leq x \leq 2, 0 \leq y \leq 2, 1 \leq z \leq 3\}$
3. $\iiint_B (x \cos y + z) dV$, where $B = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq \pi, -1 \leq z \leq 1\}$
4. $\iiint_B (z \sin x + y^2) dV$, where $B = \{(x, y, z) | 0 \leq x \leq \pi, 0 \leq y \leq 1, -1 \leq z \leq 2\}$

In the following exercises, change the order of integration by integrating first with respect to z , then x , then y .

5. $\int_0^1 \int_1^2 \int_2^3 (x^2 + \ln y + z) dx dy dz$
6. $\int_{-1}^1 \int_0^3 \int_0^1 (ze^x + 2y) dx dy dz$
7. $\int_{-1}^2 \int_1^3 \int_0^4 (x^2 z + \frac{1}{y}) dx dy dz$
8. $\int_1^2 \int_{-2}^{-1} \int_0^1 \frac{x+y}{z} dx dy dz$
9. Let F , G , and H be continuous functions on $[a, b]$, $[c, d]$, and $[e, f]$, respectively, where a, b, c, d, e , and f are real numbers such that $a < b$, $c < d$, and $e < f$. Show that

$$\int_a^b \int_c^d \int_e^f F(x)G(y)H(z) dz dy dx = \left(\int_a^b F(x) dx \right) \left(\int_c^d G(y) dy \right) \left(\int_e^f H(z) dz \right).$$

10. Let F , G , and H be differentiable functions on $[a, b]$, $[c, d]$, and $[e, f]$, respectively, where a, b, c, d, e , and f are real numbers such that $a < b$, $c < d$, and $e < f$. Show that

$$\int_a^b \int_c^d \int_e^f F'(x)G'(y)H'(z) dz dy dx = [F(b) - F(a)] [G(d) - G(c)] [H(f) - H(e)].$$

In the following exercises, evaluate the triple integrals over the bounded region $E = \{(x, y, z) | a \leq x \leq b, h_1(x) \leq y \leq h_2(x), e \leq z \leq f\}$.

11. $\iiint_E (2x + 5y + 7z) dV$, where $E = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq -x + 1, 1 \leq z \leq 2\}$
12. $\iiint_E (y \ln x + z) dV$, where $E = \{(x, y, z) | 1 \leq x \leq e, 0 \leq y \leq \ln x, 0 \leq z \leq 1\}$
13. $\iiint_E (\sin x + \sin y) dV$, where $E = \{(x, y, z) | 0 \leq x \leq \frac{\pi}{2}, -\cos x \leq y \leq \cos x, -1 \leq z \leq 1\}$
14. $\iiint_E (xy + yz + xz) dV$, where $E = \{(x, y, z) | 0 \leq x \leq 1, -x^2 \leq y \leq x^2, 0 \leq z \leq 1\}$

In the following exercises, evaluate the triple integrals over the indicated bounded region E .

15. $\iiint_E (x + 2yz) dV$, where $E = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq x, 0 \leq z \leq 5 - x - y\}$
16. $\iiint_E (x^3 + y^3 + z^3) dV$, where $E = \{(x, y, z) | 0 \leq x \leq 2, 0 \leq y \leq 2x, 0 \leq z \leq 4 - x - y\}$
17. $\iiint_E y dV$, where $E = \{(x, y, z) | -1 \leq x \leq 1, -\sqrt{1 - x^2} \leq y \leq \sqrt{1 - x^2}, 0 \leq z \leq 1 - x^2 - y^2\}$
18. $\iiint_E x dV$, where $E = \{(x, y, z) | -2 \leq x \leq 2, -\sqrt{4 - x^2} \leq y \leq \sqrt{4 - x^2}, 0 \leq z \leq 4 - x^2 - y^2\}$

In the following exercises, evaluate the triple integrals over the bounded region E of the form $E = \{(x, y, z) | g_1(y) \leq x \leq g_2(y), c \leq y \leq d, e \leq z \leq f\}$.

19. $\iiint_E x^2 dV$, where $E = \{(x, y, z) | y^2 - 1 \leq x \leq 1 - y^2, -1 \leq y \leq 1, 1 \leq z \leq 2\}$
20. $\iiint_E (\sin x + y) dV$, where $E = \{(x, y, z) | -y^4 \leq x \leq y^4, 0 \leq y \leq 2, 0 \leq z \leq 4\}$
21. $\iiint_E (x - yz) dV$, where $E = \{(x, y, z) | -y^6 \leq x \leq \sqrt{y}, 0 \leq y \leq 1, -1 \leq z \leq 1\}$
22. $\iiint_E z dV$, where $E = \{(x, y, z) | 2 - 2y \leq x \leq 2 + \sqrt{y}, 0 \leq y \leq 1, 2 \leq z \leq 3\}$

In the following exercises, evaluate the triple integrals over the bounded region

$$E = \{(x, y, z) | g_1(y) \leq x \leq g_2(y), c \leq y \leq d, u_1(x, y) \leq z \leq u_2(x, y)\}.$$

23. $\iiint_E z dV$, where $E = \{(x, y, z) | -y \leq x \leq y, 0 \leq y \leq 1, 0 \leq z \leq 1 - x^4 - y^4\}$
24. $\iiint_E (xz + 1) dV$, where $E = \{(x, y, z) | 0 \leq x \leq \sqrt{y}, 0 \leq y \leq 2, 0 \leq z \leq 1 - x^2 - y^2\}$
25. $\iiint_E (x - z) dV$, where $E = \{(x, y, z) | -\sqrt{1 - y^2} \leq x \leq 0, 0 \leq y \leq \frac{1}{2}, 0 \leq z \leq 1 - x^2 - y^2\}$
26. $\iiint_E (x + y) dV$, where $E = \{(x, y, z) | 0 \leq x \leq \sqrt{1 - y^2}, 0 \leq y \leq 1, 0 \leq z \leq 1 - x\}$

In the following exercises, evaluate the triple integrals over the bounded region

$E = \{(x, y, z) | (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$, where D is the projection of E onto the xy -plane.

27. $\iint_D (\int_1^2 (x + z) dz) dA$, where $D = \{(x, y) | x^2 + y^2 \leq 1\}$

28. $\iint_D \left(\int_1^3 x(z+1) dz \right) dA$, where $D = \{(x, y) | x^2 - y^2 \geq 1, 1 \leq x \leq \sqrt{5}\}$
29. $\iint_D \left(\int_0^{10-x-y} (x+2z) dz \right) dA$, where $D = \{(x, y) | y \geq 0, x \geq 0, x+y \leq 10\}$
30. $\iint_D \left(\int_0^{4x^2+4y^2} y dz \right) dA$, where $D = \{(x, y) | x^2 + y^2 \leq 4, y \geq 1, x \geq 0\}$
31. The solid E bounded by $y^2 + z^2 = 9$, $z = 0$, $x = 0$, and $x = 5$ is shown in the following figure. Evaluate the integral $\iiint_E z dV$ by integrating first with respect to z , then y , and then x .
[Figure: A solid arching shape that reaches its maximum along the y -axis with $z = 3$. The shape reaches zero at $y = \pm 3$, and the graph is truncated at $x = 0$ and $x = 5$.]
32. The solid E bounded by $y = \sqrt{x}$, $x = 4$, $y = 0$, $z = -2$, and $z = 1$ is given in the following figure. Evaluate the integral $\iiint_E xyz dV$ by integrating first with respect to x , then y , and then z .
[Figure: A quarter section of an oval cylinder with z from -2 to 1 . The solid is bounded by $y = 0$ and $x = 4$, and the top of the shape runs from $(0, 0, 1)$ to $(4, 2, 1)$ in a gentle arc.]
33. [T] The volume of a solid E is given by the integral $\int_{-2}^0 \int_x^0 \int_0^{x^2+y^2} dz dy dx$. Use a computer algebra system (CAS) to graph E and find its volume. Round your answer to two decimal places.
34. [T] The volume of a solid E is given by the integral $\int_{-1}^0 \int_{-x^2}^0 \int_0^{1+\sqrt{x^2+y^2}} dz dy dx$. Use a CAS to graph E and find its volume V . Round your answer to two decimal places.

In the following exercises, use two circular permutations of the variables x , y , and z to write new integrals whose values equal the value of the original integral. A circular permutation of x , y , and z is the arrangement of the variables in one of the following orders: y, z , and x or z, x , and y .

35. $\int_0^1 \int_1^3 \int_2^4 (x^2 z^2 + 1) dx dy dz$
36. $\int_1^3 \int_0^1 \int_0^{-y+1} (2y + 5z + 7x) dz dy dx$
37. $\int_0^1 \int_{-y}^y \int_0^{1-x^4-y^4} e^x dz dx dy$
38. $\int_{-1}^1 \int_0^1 \int_{-y^6}^{\sqrt{y}} (x + yz) dx dy dz$
39. Set up the integral that gives the volume of the solid E bounded by $y^2 = x^2 + z^2$ and $y = a$, where $a > 0$ and $y \geq 0$.
40. Set up the integral that gives the volume of the solid E bounded by $x = y^2 + z^2$ and $x = a^2$, where $a > 0$.
41. Find the average value of the function $f(x, y, z) = x + y + z$ over the parallelepiped determined by $x = 0, x = 1, y = 0, y = 3, z = 0$, and $z = 5$.
42. Find the average value of the function $f(x, y, z) = xyz$ over the solid $E = [0, 1] \times [0, 1] \times [0, 1]$ situated in the first octant.
43. Find the volume of the solid E that lies under the plane $x + y + z = 9$ and whose projection onto the xy -plane is bounded by $x = \sqrt{y-1}$, $x = 0$, and $x + y = 7$.

44. Find the volume of the solid E that lies under the plane $2x + y + z = 8$ and whose projection onto the xy -plane is bounded by $x = \arcsin y$, $y = 0$, and $x = \frac{\pi}{2}$.
45. Consider the pyramid with the base in the xy -plane of $[-2, 2] \times [-2, 2]$ and the vertex at the point $(0, 0, 8)$.
- Show that the equations of the planes of the lateral faces of the pyramid are $4y + z = 8$, $4y - z = -8$, $4x + z = 8$, and $-4x + z = 8$.
 - Find the volume of the pyramid.
46. Consider the pyramid with the base in the xy -plane of $[-3, 3] \times [-3, 3]$ and the vertex at the point $(0, 0, 9)$.
- Show that the equations of the planes of the side faces of the pyramid are $3x + z = 9$, $-3x + z = 9$, $-3y + z = 9$, and $3y + z = 9$.
 - Find the volume of the pyramid.
47. The solid E bounded by the sphere of equation $x^2 + y^2 + z^2 = r^2$ with $r > 0$ and located in the first octant is represented in the following figure.
[Figure: The eighth of a sphere of radius 2 with center at the origin for positive x , y , and z .]
- Write the triple integral that gives the volume of E by integrating first with respect to z , then with y , and then with x .
 - Rewrite the integral in part *a* as an equivalent integral in five other orders.
48. The solid E bounded by the equation $9x^2 + 4y^2 + z^2 = 1$ and located in the first octant is represented in the following figure.
[Figure: In the first octant, a complex shape is shown that is roughly a solid ovoid with center the origin, height 1, width 0.5, and length 0.35.]
- Write the triple integral that gives the volume of E by integrating first with respect to z , then with y , and then with x .
 - Rewrite the integral in part *a* as an equivalent integral in five other orders.
49. Find the volume of the prism with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(2, 3, 0)$, $(0, 3, 0)$, $(0, 0, 1)$, and $(2, 0, 1)$.
50. Find the volume of the prism with vertices $(0, 0, 0)$, $(4, 0, 0)$, $(4, 6, 0)$, $(0, 6, 0)$, $(0, 0, 1)$, and $(4, 0, 1)$.
51. The solid E bounded by $z = 10 - 2x - y$ and situated in the first octant is given in the following figure. Find the volume of the solid.
[Figure: A tetrahedron bounded by the xy -, yz -, and xz -planes and a triangle with vertices $(0, 0, 10)$, $(5, 0, 0)$, and $(0, 10, 0)$.]
52. The solid E bounded by $z = 1 - x^2$ and $y = 5$ and situated in the first octant is given in the following figure. Find the volume of the solid.
[Figure: A complex shape in the first octant with height 1, width 5, and length 1. The shape appears to be a slightly deformed quarter of a cylinder of radius 1 and width 5.]
53. The midpoint rule for the triple integral $\iiint_B f(x, y, z) dV$ over the rectangular solid box B is a generalization of the midpoint rule for double integrals. The region B is divided

into subboxes of equal sizes and the integral is approximated by the triple Riemann sum $\sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(\bar{x}_i, \bar{y}_j, \bar{z}_k) \Delta V$, where $(\bar{x}_i, \bar{y}_j, \bar{z}_k)$ is the center of the box B_{ijk} and ΔV is the volume of each subbox. Apply the midpoint rule to approximate $\iiint_B x^2 dV$ over the solid $B = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1\}$ by using a partition of eight cubes of equal size. Round your answer to three decimal places.

54. [T]

- (a) Apply the midpoint rule to approximate $\iiint_B e^{-x^2} dV$ over the solid $B = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1\}$ by using a partition of eight cubes of equal size. Round your answer to three decimal places.
- (b) Use a CAS to improve the above integral approximation in the case of a partition of n^3 cubes of equal size, where $n = 3, 4, \dots, 10$.

55. Suppose that the temperature in degrees Celsius at a point (x, y, z) of a solid E bounded by the coordinate planes and $x + y + z = 5$ is $T(x, y, z) = xz + 5z + 10$. Find the average temperature over the solid.

56. Suppose that the temperature in degrees Fahrenheit at a point (x, y, z) of a solid E bounded by the coordinate planes and $x + y + z = 5$ is $T(x, y, z) = x + y + xy$. Find the average temperature over the solid.

57. Show that the volume of a right square pyramid of height h and side length a is $v = \frac{ha^2}{3}$ by using triple integrals.

58. Show that the volume of a regular right hexagonal prism of edge length a is $\frac{3a^3\sqrt{3}}{2}$ by using triple integrals.

59. Show that the volume of a regular right hexagonal pyramid of edge length a is $\frac{a^3\sqrt{3}}{2}$ by using triple integrals.

60. If the charge density at an arbitrary point (x, y, z) of a solid E is given by the function $\rho(x, y, z)$, then the total charge inside the solid is defined as the triple integral $\iiint_E \rho(x, y, z) dV$.

Assume that the charge density of the solid E enclosed by the paraboloids $x = 5 - y^2 - z^2$ and $x = y^2 + z^2 - 5$ is equal to the distance from an arbitrary point of E to the origin. Set up the integral that gives the total charge inside the solid E .